

TRAYA YUKTI (TYC) AI LMS

Next-Generation AI-Powered Learning Management System & Developer Ecosystem

Version: 2.5.0 Production Ready

Stack: React 19, TypeScript, Vite, Tailwind CSS

Auth: 6-Phase Mascot + Clerk Enterprise

1. Executive Summary & Core Mission

Traya Yukti Core (TYC) LMS is a state-of-the-art educational ecosystem architected to bridge academic learning and production software engineering. Built under the mantra '**Learn • Build • Grow**', the platform combines interactive video courses, browser-based coding labs, portfolio generation, capstone project reviews, and context-aware generative AI mentoring.

Key value drivers of the platform include zero-friction authentication with a 3D responsive character state machine, granular Role-Based Access Control (RBAC), multi-track learning roadmaps, real-time gamified skill trees, and full administrative curriculum CMS management.

2. Interactive Character Authentication & Security

The TYC authentication system features a master 3D mascot interface that dynamically reacts to user actions across six distinct interaction phases:

Phase	Trigger / State	Character Visual Reaction	Security & System Action
State 1	Email Focus / Input	Character looks towards input with ■ badge	Real-time email format regex evaluation active
State 2	Valid Email Format	Thumbs-up approval with green ✓ badge	Green glowing border & checkmark displayed
State 3	Invalid Email / Mistake	Confused chin-scratch with purple ? badge	Red border, error alert, access strictly gated
State 4	Password Typing	Character covers eyes with ■ privacy badge	Client-side password masking & privacy guard
State 5	Valid Password (≥6)	Character smiles, looks back with ■ badge	Password policy verified, submit button unlocked
State 6	Login Success	Hands-raised celebration with ■ neon badge	Confetti burst, progress bar, redirect to /dashboard

Clerk Enterprise Authentication: In addition to local RBAC accounts, the platform integrates production-grade Clerk Authentication (@clerk/react) linked to application app_3lilajRXTOubxyGOwlo5qDsQoAh, offering Google OAuth, biometric authentication, and multi-factor security.

3. Core Learning Modules & Functional Features

- **Course Catalog & Video Player:** Comprehensive multi-tier courses (Full Stack React 19, Python AI Backend, Cloud DevOps, Prompt Engineering) with video streaming, lesson notes, code snippets, and automated progress checkpointing.

- **Guided Learning Paths:** Curated end-to-end career tracks guiding students from foundational concepts to advanced production deployments with milestone verification.
- **Interactive Coding Lab & Practice:** In-browser code editor supporting JavaScript, TypeScript, Python, and SQL with instant execution, test runner, output terminal, and algorithmic problem sets.
- **Capstone Projects & Submissions:** Industry-grade projects requiring architecture design, GitHub repository submissions, live preview links, and rubric-based instructor grading.
- **Verifiable Certificates & Portfolio:** Automated certificate generation upon 100% course completion with unique verification hashes, LinkedIn sharing, and a personalized public student portfolio generator.
- **24/7 Floating AI Tutor (TYC Assistant):** Interactive drawer equipped with four specialized AI modes: Code Debugger, Architecture Explainer, Quiz Master, and Career Mentor.
- **Community, Workshops & Hackathons:** Collaborative forums, live workshop schedules, and hackathon registration portals fostering peer-to-peer developer collaboration.

4. Role-Based Access Control (RBAC) Matrix

Role	Default User	Password	Clearance & Accessible Portals
Student	alex.rivera@tyc.dev	alex@123	Student Dashboard, Courses, Coding Lab, Practice, Portfolio
Student	priya.sharma@tyc.dev	priya@123	Student Dashboard, Cloud DevOps Tracks, Certificates
Instructor	sarah.chen@tyc.dev	sarah@123	Instructor Portal, Grading Submissions, Student Roster
Admin	admin@tyc.dev	admin@123	Admin Console, Course CMS, Student Manager, System Analytics
Owner (Creator)	owner@tyc.dev	owner@123	Unrestricted root clearance across all consoles & CMS features

5. Technical Architecture & Component Tree

Layer	Technology / Library	Architectural Purpose
Core Framework	React 19 + TypeScript	Strict typing, concurrent rendering, modular UI chunks
Build Tooling	Vite 8	Sub-second Hot Module Replacement (HMR) and optimized build chunks
Styling & Theme	Tailwind CSS + CSS Tokens	High-contrast Dark/Light themes, responsive typography, glassmorphism
Animation Engine	Framer Motion + Canvas Confetti	Smooth Mascot crossfades, route transitions, celebration particles
Enterprise Auth	@clerk/react + LocalStorage RBAC	Multi-provider authentication, OAuth, secure session persistence
State Contexts	Auth, LMS, Notifications, Theme	Decoupled global state management with zero external boilerplate

6. Platform Verification & Running Instructions

To access and test the entire platform locally:

1. **Start Server:** Run `npm run dev` or `vite` (Server runs at `http://localhost:5173/`).
2. **Access Login Gate:** Navigate to `http://localhost:5173/login` to experience the 6-phase mascot animation.
3. **Sign In:** Use any of the RBAC credentials above or 1-Click Google / Clerk Auth to unlock the dashboard.
4. **CMS & Admin Console:** Log in as `admin@tyc.dev` or `owner@tyc.dev` to manage courses, students, and submissions.

■ **Summary Statement:** *Traya Yukti Core (TYC) LMS delivers an enterprise-grade, pedagogically sound, and visually engaging developer education ecosystem ready for real-world deployment.*